



Nitrogen-induced hardening of refractory high entropy alloys containing laminar ordered phases

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ABSTRACT

Most attempts to improve the properties of high entropy alloys (HEAs) involve the exploration of non-equimolar compositions, addition of alloying elements, and/or manipulation of the microstructure. Alternatively, this work reports on intentionally doping HEAs with non-metallic species to precipitate coherent, ordered phases in order to maximize hardness. A refractory MoNbTaW HEA was synthesized via cryogenic mechanical alloying and doped with nitrogen by using liquid nitrogen as the cryogen. Overall, the doping strategy was successful as multiple nitrogen-rich secondary phases were observed. In particular, a unique ordered laminar phase was developed and identified as tetragonal (Mo,W)(Nb,Ta)N nitride via aberration-corrected scanning transmission electron microscopy (STEM). Another complex (Nb,Ta)₂CN carbonitride was also identified. The growth behaviors of the ceramic phases were studied using long term aging treatments of up to 100 hours at 1200°C. A second MoNbTaW alloy was also prepared via high-energy ball milling, without nitrogen, by using liquid Ar as the cryogen. A comparison of the two alloys' microstructures and properties confirm that intentional formation of complex nitride phases greatly enhanced the hardness of mechanically alloyed MoNbTaW by up to 3–4 GPa.

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1. Introduction

There are currently many ongoing efforts to develop the next generation of advanced materials that could potentially replace Ni-based super alloys for extreme applications. One class of materials that show promise are refractory high entropy alloys (RHEAs) [1–3]. Comprised of four or more refractory elements (e.g. Hf, Mo, Nb, Ta, Ti, V, W, Zr) ranging in concentrations between 5–35 at.%, RHEAs have been shown to exhibit strengths in excess of 400 MPa at temperatures up to 1600°C [4–6]. Notwithstanding, considerable efforts have been made to further enhance the mechanical properties of RHEAs by incorporating additional alloying elements [4,6–12] and/or applying complex processing and/or deformation techniques to modify microstructures [13–22].

A specific approach to improve the thermomechanical performance of HEAs is precipitation strengthening via incoherent and/or coherent ordered phases, similar to L1₂-γ' Ni₃(Al,Ti) precipitates in Ni-based super alloys [23–27], θ and β precipitates in Al alloys [28–32], and nanoscale oxides in oxide dispersion

strengthened (ODS) alloys [33–35]. Various precipitation strengthened HEAs have already been synthesized, although most examples involve transition metal HEAs that rely on the formation of secondary metallic solid solutions or intermetallics [36,37]. For instance, additions of Al and Ti in Ni-based HEAs form ordered L1₂ and B2 phases that improve hardness [38] or strength and ductility [39–41]. Overall, while transition metal HEAs can be precipitation strengthened by, for example, imitating γ-γ' microstructures, similar success has not been demonstrated in RHEAs.

Considering the difficulties in controlling coherency of metallic phases in RHEAs, an alternative design strategy to improve mechanical performance is to intentionally precipitate ordered ceramic phases by doping with non-metallic elements (e.g., C, N, O). The viability of incorporating non-metallic elements (e.g. C, N, O) into RHEAs has previously been demonstrated by arc-melting and/or mechanical alloying [21,42–44]. Oxygen, in particular, has been shown to form ordered oxygen complexes that enhance strength and ductility [42,43]. While no evidence of ordered nitrides was observed in these experiments, one particular experiment, in which cryogenic mechanical alloying was employed, reported on the formation of various complex nitrides which precipitated upon annealing at 1000°C. These nitride phases were

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shown to improve the overall hardness of the alloy, contributing to an increase in hardness of almost 400% as compared to other MoNbTaW alloys which were processed using more conventional methods (e.g., casting) [21]. On a similar note, mechanical alloying of binary nitride precursors has recently been shown to produce bulk high entropy nitrides that exhibit high hardness [45]. One fall-back of bulk high entropy nitrides, however, is the loss of ductility that is ideal for applications that often require Ni-based superalloys. Therefore, there is a need to develop high entropy materials that exhibit ultrahigh hardness (e.g., high entropy nitrides) while also potentially maintaining reasonable ductility (e.g., high entropy metals).

This work aimed to develop a nanocrystalline composite consisting of high entropy refractory metallic phases combined with ordered complex nitride phases that achieves superior mechanical properties. A secondary goal was to evaluate the atomic strengthening mechanisms associated with ordered nitride phases. Therefore, two distinct alloys were synthesized via cryogenic mechanical alloying. One material was milled using liquid nitrogen to incorporate N and induce nitride formation, while the second alloy was milled with liquid argon to minimize N additions for comparison. Both alloys were then subjected to long term annealing treatments to evaluate the formation and growth behavior of complex nitrides. Microstructures were characterized via aberration-corrected scanning transmission electron microscopy (STEM). Vicker's microindentation was performed to measure the effect of the complex nitrides on hardness and ultimately to determine the optimal heat treatment. Overall, this study reports on the effect of successful nitrogen doping on the formation of coherent, nitride phases and their effect on the mechanical properties of mechanically alloyed MoNbTaW. Furthermore, the stability, microstructural characteristics, and strengthening mechanisms of the coherent, ordered nitride phase will be discussed.

2. Materials and Methods

2.1. Bulk Sample Preparation

Two refractory HEAs with intended compositions of $\text{Mo}_{25}\text{Nb}_{25}\text{Ta}_{25}\text{W}_{25}$ were synthesized via high-energy cryogenic mechanical alloying. High purity elemental powders (Alfa Aesar, 99.9 metals basis) were loaded into tool steel SPEX mixing vials with tungsten carbide ball bearings. The ball to powder ratio was 10:1. A modified SPEX 8000M mill was used to mechanically alloy the powders for 8 hours at cryogenic temperatures. One alloy was milled using liquid nitrogen to intentionally introduce N while the second alloy was milled using liquid argon to minimize incorporation of N. The two alloys will hereafter be referred to as HEA-N₂ and HEA-Ar. Once mixed, 3 mm compacts were produced by uniaxially pressing loose powder using a tungsten carbide die with an applied force of 3 GPa. Lastly, compacts were sealed in quartz ampules backfilled with Ar gas and annealed at 1000°C for 1, 100, and 1000 hours or 1200°C for 100 hours. Samples were furnace cooled at a rate of ~3°C/min.

2.2. Hardness Testing

Vicker's microhardness testing was completed on as-milled and annealed compacts using a Buehler Tukon 1202 indenter. At least 15 indents were taken from polished surfaces using a load of 50 grams and dwell time of 10 seconds. Indents were applied in the center of the largest particles to minimize contributions from porous areas.

2.3. Microstructural Characterization

Quantitative x-ray energy dispersive spectroscopy (EDS) via scanning electron microscopy was conducted to determine concentrations of metallic species. A FEI Scios dual-beam focused ion beam (FIB) instrument was operated at 30 kV and equipped with an octane elite silicon drift detector (SDD) to collect a minimum of 20 spectra from polished compacts. Quantitative x-ray analysis was performed using an approach described by Newbury et al. [46–48]. An additional MoNbTaW alloy with a known composition, confirmed by Luvak Laboratories Inc., was used as a reference material. Non-metallic species concentrations were measured by Luvak Laboratories, Inc. via combustion infrared detection (ASTM E 1019-18) and inert gas fusion (ASTM E 1019-18 & ASTM E 1447-16). Ar concentration was not quantified due to detectability limitations.

Backscattered electron images of polished, as-milled and annealed compacts were acquired using the FIB instrument operated at 10 kV. Thin specimens for aberration-corrected scanning transmission electron microscopy (STEM) were prepared using FIB in-situ lift-out methods [49,50]. Specimens were mounted on either Mo or Si grids (Ted Pella). Primary extraction and thinning were performed with a 30 kV Ga ion beam. The Ga ion voltage was reduced to 5 kV during the final thinning steps to reduce ion beam damage. A Fischione 1040 NanoMill with an Ar ion beam operated at 900 eV was used as a final milling processes to remove Ga implantation. STEM was conducted using a JEOL ARM-200CF instrument operated at 200 kV. X-ray energy dispersive spectroscopy (EDS) maps were collected using a JEOL Centurio 100 mm² silicon drift detector. Bright field (BF), high-angle annular dark field (HAADF), and low-angle annular dark field (LAADF) images were simultaneously collected. Fast Fourier transforms (FFTs) and intensity profiles were extracted from STEM HAADF images to facilitate phase identification. JEMS microscopy simulation software was used to simulate, match, and index the FFTs. Lastly, Visualization for Electronic and Structural Analysis (VESTA) software was used to interpret atomic structures.

3. Results

3.1. Bulk Composition and Microstructure

The metallic and non-metallic concentrations in each as-milled alloy are compared in Table 1. Both alloys, having intended compositions of $\text{Mo}_{25}\text{Nb}_{25}\text{Ta}_{25}\text{W}_{25}$, exhibited increased amounts of tungsten and carbon contamination likely as a result of being milled with WC media. Both alloys also showed minor amounts of Fe contamination (~4at%) originating from the walls of the stainless-steel vial. The greatest difference between the two alloys was the concentration of nitrogen: HEA-N₂ consisted of ~24 at.% N while HEA-Ar consisted of ~1 at.% N. More detailed metallic and non-metallic concentrations are listed in Supplemental Table 1.

A comparison of HEA-N₂ and HEA-Ar compacts in the as-milled and annealed conditions at 1200°C for 100 hours is shown in the backscattered electron micrographs of Fig. 1. Fig. 1a and 1c indicate both as-milled alloys achieved a nearly homogenous mixture of all elemental powders. Fine (<500 nm) particles having a higher intensity than the solid solution, most likely WC chips, can be seen in both as-milled compacts. Limited amounts of porosity were also observed as marked by regions with the lowest intensity. Fig. 1b and d show that, after annealing at 1200°C for 100 hours, each alloy exhibited a multiphase nanocrystalline microstructure. Most notably, the two alloys exhibited strikingly different grain morphologies; HEA-N₂ exhibited elongated grains with intensity variations of nearly 300 nm in width comprising nearly half the mi-

Table 1

Primary (Nb, Mo, Ta, W) and secondary (Fe, C, N) atomic concentrations of each alloy in the as-milled condition. Metallic concentrations were determined via quantitative X-ray energy dispersive spectroscopy (EDS) and non-metal concentrations were determined via combustion and inert gas fusion techniques.

Alloy	Principal Elements (at %)				Secondary Elements (at %)			
	Nb	Mo	Ta	W	Fe	C	N	Other
HEA-N ₂	13.1 ± 0.3	13.1 ± 0.5	12.7 ± 0.2	21.2 ± 0.3	3.6 ± 0.2	7.8	23.8	4.7
HEA-Ar	20.2 ± 0.4	21.7 ± 0.7	21.1 ± 0.4	25.0 ± 0.5	4.7 ± 0.2	2.1	0.9	4.3

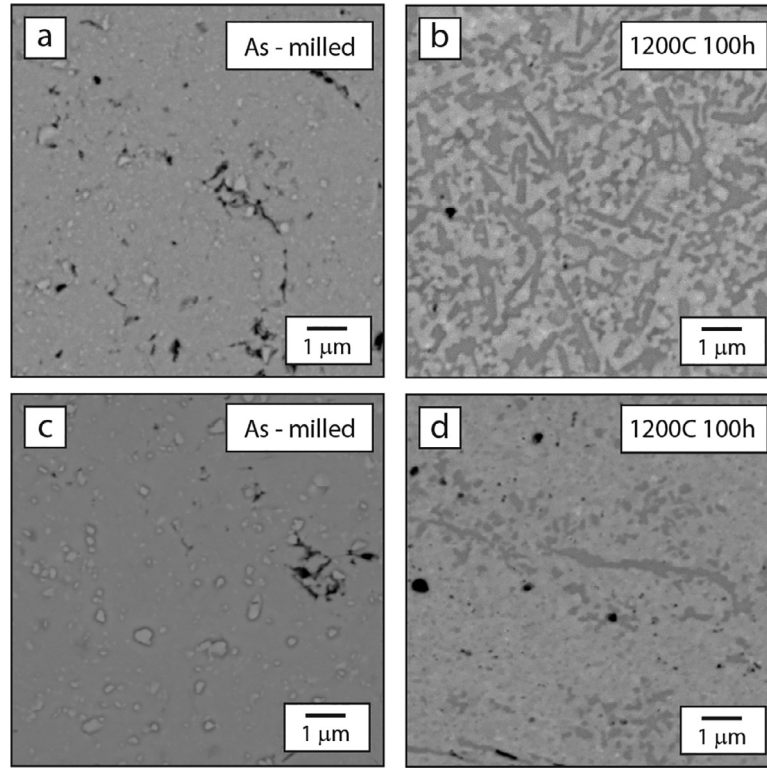


Fig. 1. Backscattered electron images of uniaxially pressed compacts of (a,b) HEA-N₂ and (c,d) HEA-Ar in the as-milled and 1200°C 100 hour annealed conditions. Bulk microstructures in the as-milled condition were similar, but differed after annealing with HEA-N₂ exhibiting a more pronounced phase separation.

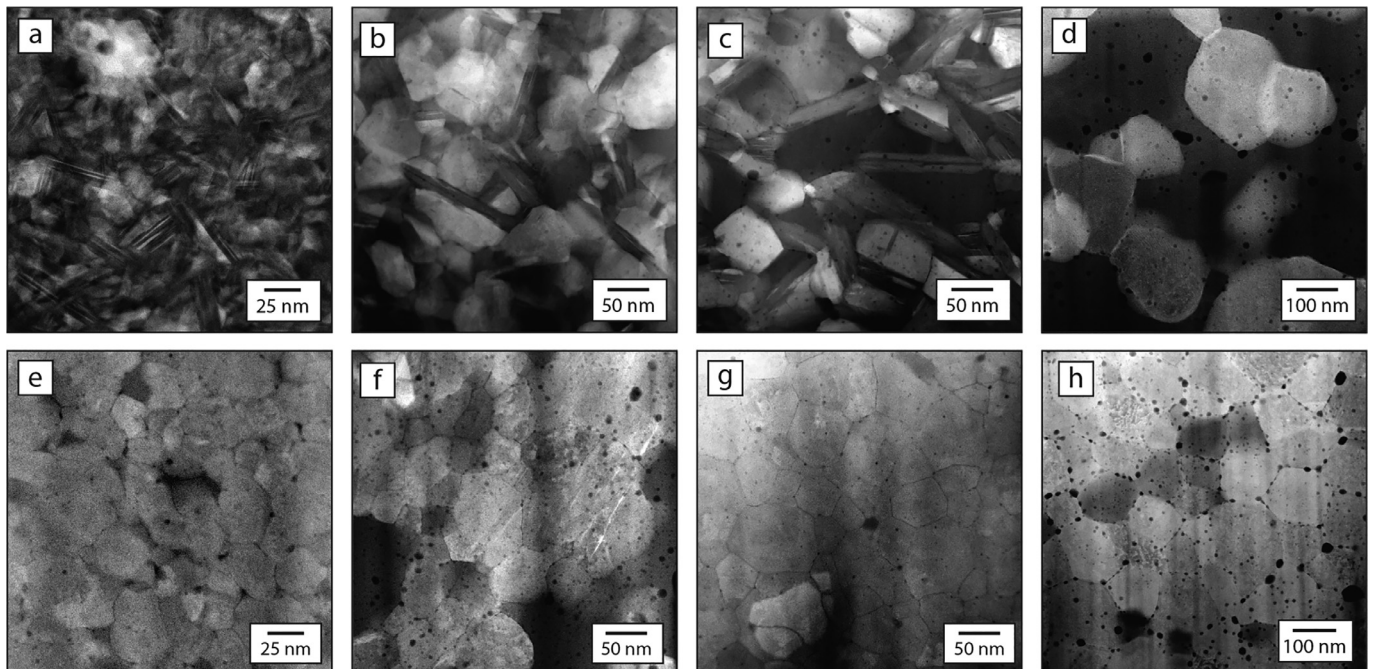


Fig. 2. STEM-HAADF images of HEA-N₂ (a-d) and HEA-Ar (e-h) after annealing at 1000°C for 1, 100, and 1000 hours, and 1200°C for 100 hours, respectively.

crostructure, while HEA-Ar contained a smaller area fraction of a darker phase that exhibited mostly equiaxed grains (Fig. 1d).

Due to resolution limitations in backscatter imaging, aberration-corrected STEM imaging was carried out to better resolve microstructural differences between the alloys. First, both as-milled alloys exhibited mostly amorphous microstructures with 5–10 nm diameter scattered crystallites (Supplemental Figure 1). Next, STEM-HAADF micrographs of HEA-N₂ and HEA-Ar in each annealing condition are shown in Fig. 2. Similar to Fig. 1, Fig. 2a and e show that HEA-N₂ and HEA-Ar, respectively, comprised of multiphase nanocrystalline microstructures with a median grain size of ~25 nm after annealing at 1000°C for 1 hour. Once again, HEA-N₂ contained elongated grain morphologies. Differences in the grain morphology are more clearly discernable after annealing for longer times. Large populations of elongated, lath-like grains were consistently observed in HEA-N₂ after annealing at 1000°C (Fig. 2a–c). However, these lath-like grains in HEA-N₂ were not present after annealing at 1200°C for 100 hours (Fig. 2d). Equiaxed grains were consistently observed in all the annealed HEA-Ar alloys (Fig. 2e–h). In all cases, HEA-N₂ consisted of a ~50/50 mixture of phases whereas HEA-Ar was mostly single phase with ~20% area coverage of the darker phase. The reduced HAADF intensity of the darker phases suggests that they have lower average atomic number due to the mass thickness contrast mechanisms of HAADF imaging. Thus, as an example, the brighter phase may be enriched with W while the darker phases may be enriched with Nb and/or non-metallic species (e.g., C and N). Dark spherical features were also observed in each alloy and later determined to be Ar bubbles that were trapped during back-filling of vials with Ar prior to milling [51]. Finally, while grain growth was observed, both alloys remained nanocrystalline after annealing for 100 hours at 1200°C.

STEM-EDS was conducted to determine the relative compositions of each phase in both HEA-N₂ and HEA-Ar. Fig. 3a and b include STEM HAADF reference images taken from HEA-N₂ and HEA-Ar after annealing at 1000°C for 1000 hours as well as X-ray intensity maps of C K α , N K α , Nb K α , Mo K α , Ta L α , and W L α x-ray peaks. This annealing condition was chosen for further analysis because the grains were the largest (minimize overlapping grains) and all the phases are still present in HEA-N₂ (the lath-like grains were not observed after annealing at 1200°C). The X-ray maps generally demonstrate that Nb and Ta were correlated. Similarly, Mo and W were correlated. Fig. 3a also indicates that HEA-N₂ contains a Mo-W-rich phase (labeled A) and two different Nb-Ta-rich phases: an equiaxed phase (labeled B) enriched with C-N-Nb-Ta and an elongated phase with planar variations of high and low contrast (labeled C) enriched with a relatively higher amount of N. Thus, it was determined that HEA-N₂ contained a complex ceramic phase containing nitrogen and carbon (i.e., a carbonitride) as well as a complex nitride phase (i.e., without carbon). Lastly, Fig. 3b indicates that HEA-Ar was mostly comprised of a Nb-Mo-Ta-W-rich phase and minor amounts of a Nb-Ta-C-rich phase (i.e., carbide phase).

3.2. Complex Ceramic Phase Identification

The lath-like nitride phase and equiaxed carbonitride phase observed after annealing at 1000°C for 1000 hours were further identified using atomic-resolution STEM-HAADF imaging. The phases were identified by collecting multiple images on different zone axes. Fig. 4 first shows examples of STEM-HAADF images of the lath-like nitride phase. Fig. 4a and e show images taken from the [100] and [001] crystal directions, respectively. Fig. 4b and f are the corresponding fast Fourier transforms (FFTs) which can be treated similarly to selected area diffraction patterns (SADP). Here it was determined that the lath-like nitride phase was tetragonal with measured lattice parameters of $a = b = 3.096 \text{ \AA}$ and $c = 7.843 \text{ \AA}$.

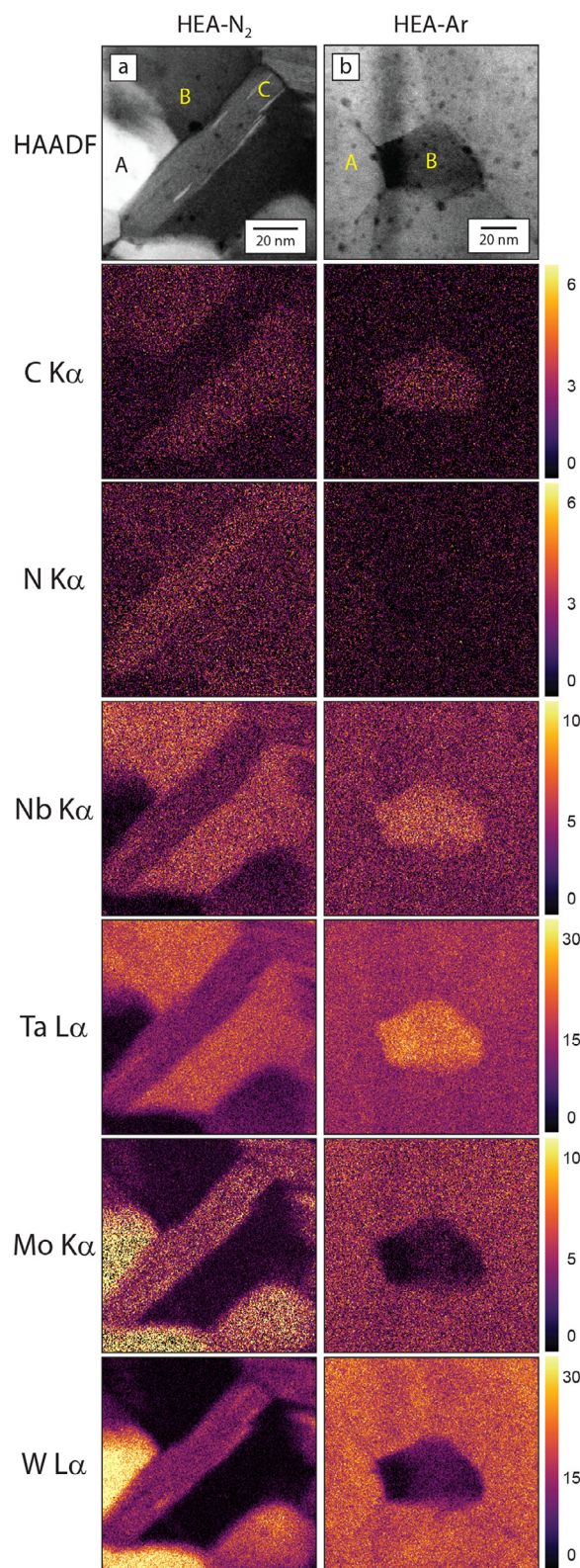


Fig. 3. STEM-EDS spectrum image of microstructure of (a) HEA-N₂ and (b) HEA-Ar annealed at 1000°C for 1000 hours. Each data set includes a reference HAADF micrograph and X-ray intensity maps of C K α , N K α , Nb K α , Mo K α , Ta L α , and W L α . The color scale represents integrated counts per pixel.

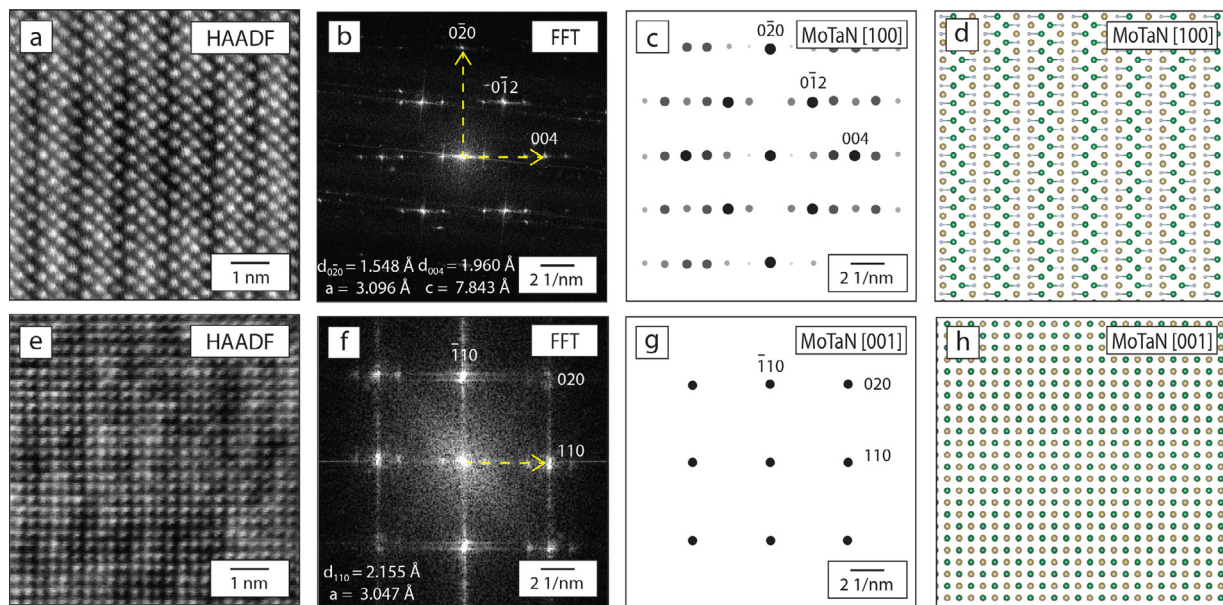


Fig. 4. (a,e) STEM-HAADF images and corresponding (b,f) Fast Fourier Transforms (FFTs) taken from the [100] and [001] crystallographic directions of the lath-like nitride phase, respectively. The experimental FFTs (b,f) are consistent with simulated diffraction patterns from (c) the [100] and (g) the [001] crystal directions of tetragonal MoTaN, respectively. VESTA atomic structures of MoTaN down the [100] and [001] directions are shown in (d) and (h), respectively, where the gold atoms are Ta, green atoms are Mo, and purple atoms are N. The HAADF micrographs were acquired from HEA-N₂ annealed at 1000°C for 1000 hours.

The measured d-spacings, crystallographic symmetry, and compositional results from Fig. 3 were used to determine potential candidate Nb-Ta-rich nitrides. Based on the measured compositions and lattice parameters, the ternary nitride MoTaN was identified as a candidate phase. Simulated diffraction patterns from the [100] and [001] crystal directions MoTaN are shown in Fig. 4c and g, respectively, showing good agreement with the FFTs shown in Fig. 4b and f. VESTA atomic projections of the MoTaN [100] and [001] crystal directions are shown in Fig. 4d and h, respectively to unequivocally confirm that the observed complex nitride phase matched that of MoTaN. The gold, green, and purple atoms represent Ta, Mo, and N atoms, respectively. Nitrogen atoms are included in the VESTA insets, but they are not distinguishable in these STEM HAADF images. In summary, the lath-like phase was determined to be structurally similar to tetragonal MoTaN (space group P4/nmm), having a reported lattice parameter of $a = b = 3.095$ Å and $c = 7.866$ Å [52,53].

As demonstrated in Fig. 4d, the tetragonal MoTaN phase consists of alternating layers enriched with Mo and Ta. Considering that Nb and W were observed in the phase, according to Fig. 3a, further characterization was conducted via STEM EDS to correlate the changes in HAADF intensity with the spatial distribution of elements in the sublattice. Fig. 5 shows a spectral image of the layered lattice structure in HEA-N₂ annealed at 1000°C for 1000 hours. The STEM HAADF reference image is shown in Fig. 5a. Since Fig. 3 demonstrated that Mo was correlated with W and Nb was correlated with Ta, Fig. 5b and c show X-ray intensity maps of Mo $K\alpha$ + W $L\alpha$ and Nb $K\alpha$ + Ta $L\alpha$, respectively. The N $K\alpha$ X-ray intensities were not strong enough to plot. Representative line profiles of HAADF intensity and X-ray intensities, from left to right, are also shown below the X-ray maps. The X-ray line profiles were averaged every two pixels to reduce noise. Overall, the maximum HAADF intensities correlated with Mo $K\alpha$ + W $L\alpha$ X-ray signals while the reduced HAADF intensities correlated with Nb $K\alpha$ and Ta $L\alpha$ X-ray signals. The STEM-EDS spectral image further confirmed that the crystal structure of the lath-like nitride phase was similar to MoTaN considering that alternating compositional layers were

observed via HAADF imaging and STEM-EDS. The results further indicate that the MoTaN observed here contained additional elements such as W and Nb. In this case, the results indicate that Mo and W occupy the same lattice sites while Nb and Ta occupy the same lattice sites. Therefore, based on the STEM HAADF imaging and STEM-EDS results shown in Figs. 4 and 5, the lath-like nitride phase was fully identified as (Mo,W)(Nb,Ta)N.

A similar approach using STEM-HAADF imaging was used to determine the crystallography of the equiaxed C-N-Nb-Ta-rich carbonitride phase. The results are shown in Fig. 6. Based on the FFT analysis shown in Fig. 6, it was determined that the phase was cubic with a measured lattice parameter of $a = 4.430$ Å. Fig. 6a-d show the [001] crystal direction while Fig. 6e-h show the [110] crystal direction. Considering this phase was rich in C-N-Nb-Ta, and that Nb and Ta are known to form carbonitrides exhibiting cubic rock salt structures [54], the carbonitride, Nb₂CN, was considered as the primary candidate for the identification of this phase. Simulated diffraction patterns for the [001] and [110] crystal directions of NbN, a similar rock salt cubic crystal structure, are shown in Fig. 6c and g, respectively. The patterns are in good agreement with the FFTs of the equiaxed phase shown in Fig. 6b and f. It was therefore determined that the equiaxed C-N-Nb-Ta-rich carbonitride phase was structurally similar to the rock salt NbN crystal structure having a reported lattice parameter of $a = b = c = 4.44$ Å [55]. Fig. 6d and h show VESTA projections down the [001] and [110] crystallographic directions of NbN further confirming that the crystal structure of the Nb-Ta carbonitride phase was structurally similar to NbN. The red and purple atoms represent Nb and N atoms, respectively. The STEM-EDS map shown in Fig. 3a indicated that in addition to Nb, C, and N, the equiaxed carbonitride phase comprised of Ta. It was therefore determined that the Nb and Ta atoms share lattice sites and thus the phase was identified as (Nb,Ta)₂CN. Similar to the ceramic phases found in HEA-N₂, the darker phase observed in all annealing conditions of HEA-Ar was identified as (Nb,Ta)₂C using a STEM-EDS map, shown in Fig. 3b, and STEM HAADF images. The crystallographic analysis is shown in Supplemental Figure 2.

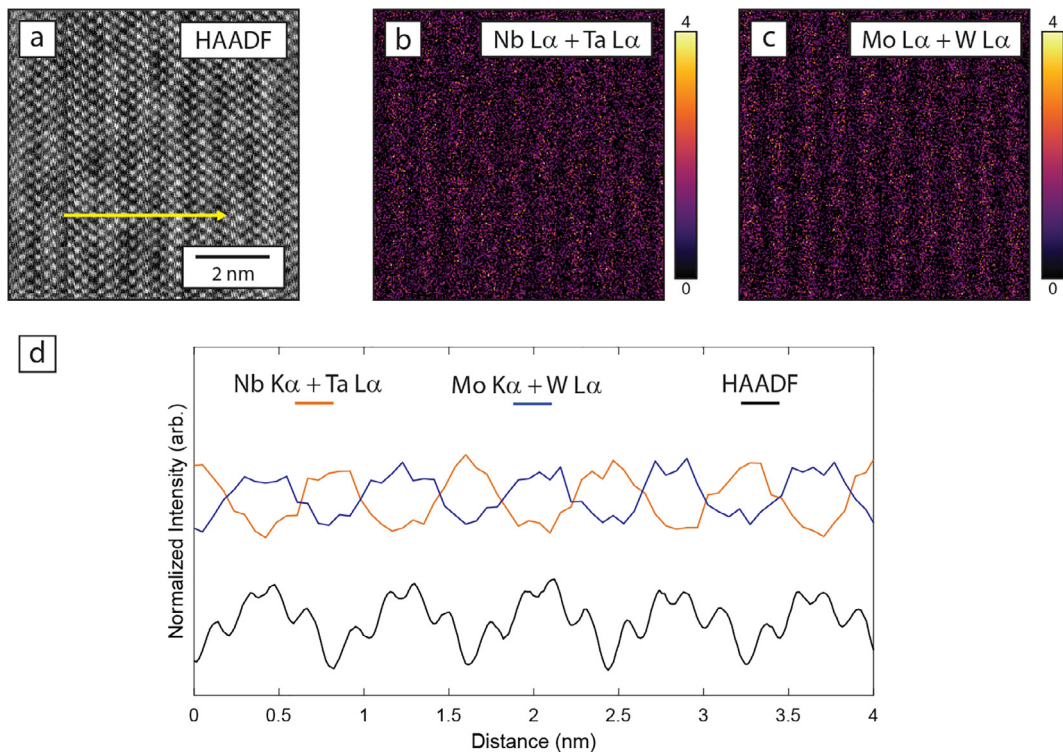


Fig. 5. STEM-EDS spectrum image of layered lattice structure in HEA-N₂ annealed at 1000°C for 1000 hours: (a) HAADF reference image, (b) combined Mo K α + W L α X-ray maps, (c) combined Nb K α + Ta L α X-ray maps, and (d) integrated profiles of HAADF intensity and X-ray signals from left to right.

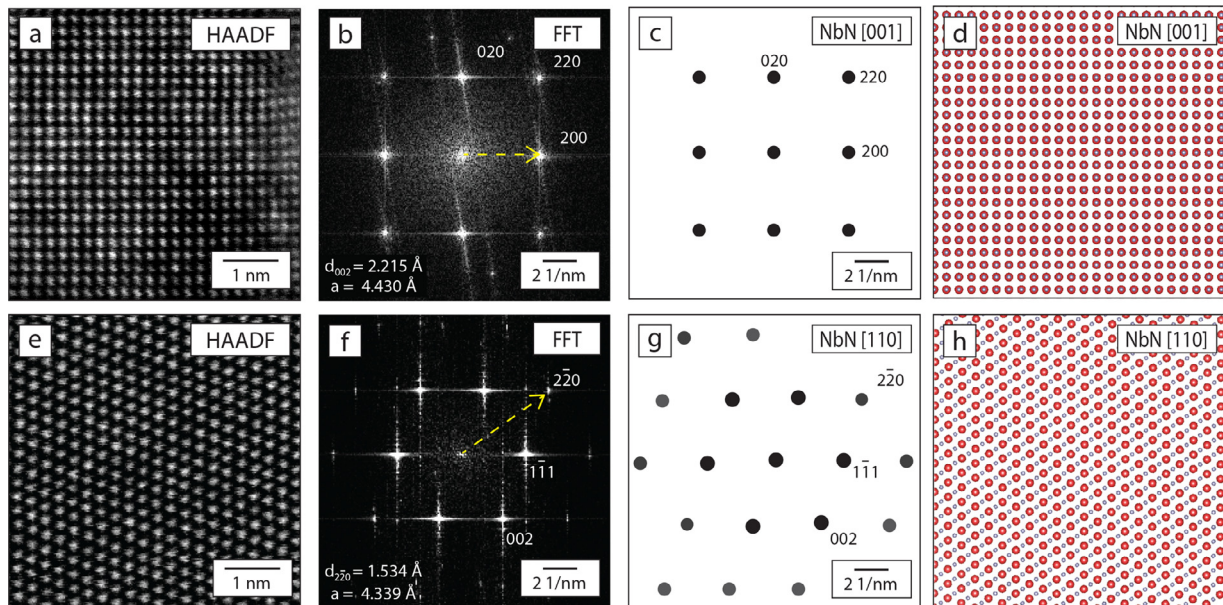


Fig. 6. (a,e) STEM-HAADF images and corresponding (b,f) Fast Fourier Transforms (FFTs) taken from the [001] and [110] crystallographic directions of the equiaxed nitride phase, respectively. The experimental FFTs (b,f) are consistent with simulated diffraction patterns from (c) the [001] and (g) the [110] crystal directions of cubic NbN, respectively. VESTA atomic structures of NbN down the [001] and [110] directions are shown in (d) and (h), respectively, where the red atoms are Nb and purple atoms are N. The HAADF micrographs were acquired from HEA-N₂ annealed at 1000°C for 1000 hours.

3.3. Determination of Orientation Relationships

The changes in HAADF contrast within individual grains in HEA-N₂ after annealing at 1000°C, as shown most clearly in Fig. 2a-c, suggest that there are changes in composition inside particular grains. Therefore, it is likely that a separate phase(s) precipitated and/or grew within primary grains. A first example from the HEA-N₂ alloy annealed at 1000°C for 1 hour is shown in Fig. 7. Here is

it clear that a single grain is composed of both the (Nb,Ta)₂CN and (Mo,W)(Nb,Ta)N carbonitride and nitride phases. One possibility is that (Mo,W)(Nb,Ta)N layers grew in a ledge-like step growth fashion (denoted by yellow arrows), protruding from the grain exterior into the center of the grain, all while maintaining coherency with the parent (Nb,Ta)₂CN grain. Fast Fourier Transforms (FFTs) were extracted from both the (Nb,Ta)₂CN region (white dashed box) and (Mo,W)(Nb,Ta)N region (yellow dashed box) to determine their

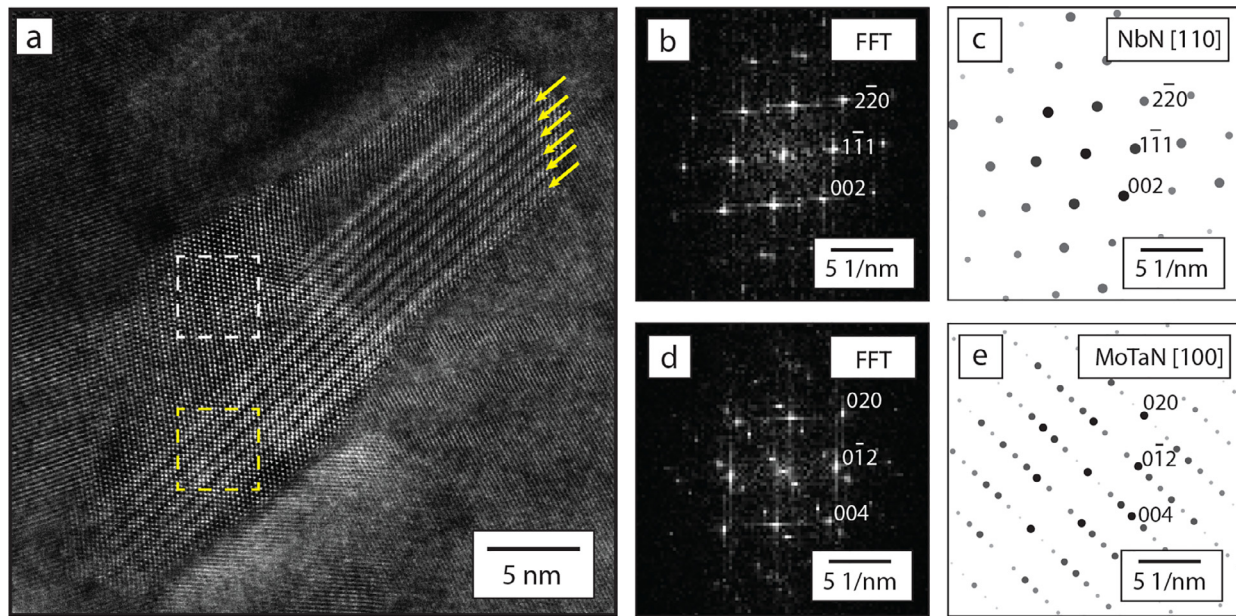


Fig. 7. (a) STEM-HAADF image showing orientation relationship between $(\text{Nb,Ta})_2\text{CN}$ and $(\text{Mo,W})(\text{Nb,Ta})\text{N}$. The white and yellow dashed boxes mark regions of interest in the $(\text{Nb,Ta})_2\text{CN}$ and $(\text{Mo,W})(\text{Nb,Ta})\text{N}$ phases, respectively. Corresponding Fast Fourier Transforms (FFTs) taken from $(\text{Nb,Ta})_2\text{CN}$ and $(\text{Mo,W})(\text{Nb,Ta})\text{N}$ are compared in (b) and (d), respectively. The FFTs are consistent with indexed diffraction patterns from the (c) [110] and (e) [100] crystal directions of cubic NbN and tetragonal MoTaN, respectively.

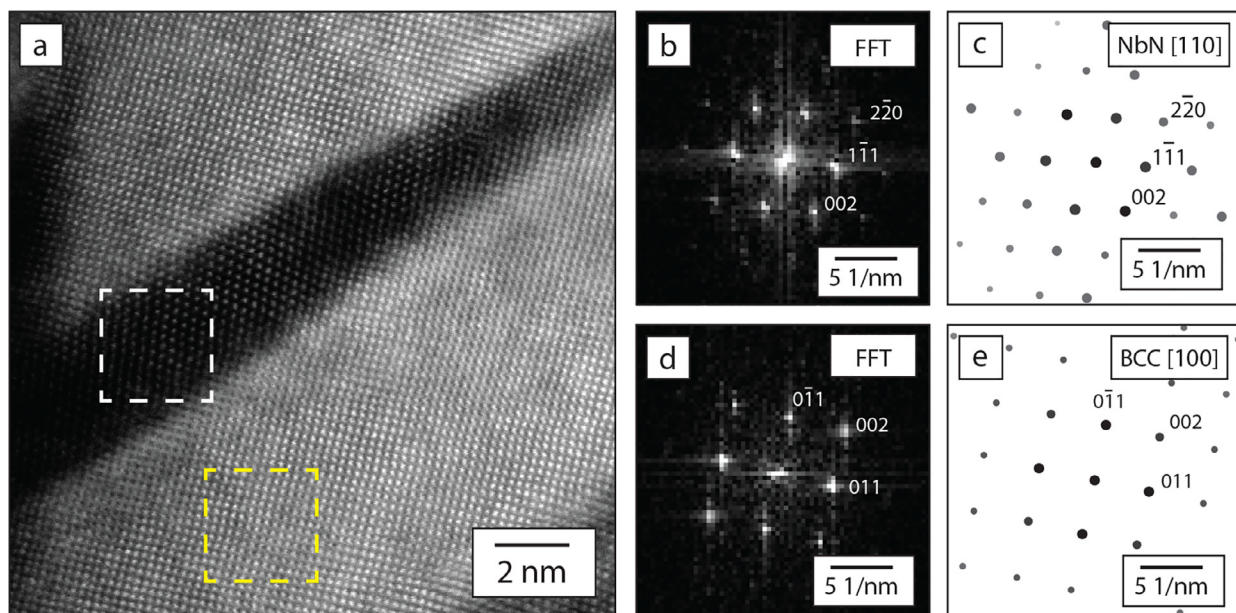


Fig. 8. (a) STEM-HAADF image showing orientation relationship between $(\text{Nb,Ta})_2\text{CN}$ and the BCC HEA phase. The white and yellow dashed boxes mark regions of interest in the $(\text{Nb,Ta})_2\text{CN}$ and BCC phases, respectively. Corresponding Fast Fourier Transforms (FFTs) taken from $(\text{Nb,Ta})_2\text{CN}$ and the BCC phase are compared in (b) and (d), respectively. The FFTs are consistent with indexed diffraction patterns from the (c) [110] and (e) [100] crystal directions of cubic NbN and BCC, respectively.

crystallographic relationship. The crystallographic relationship between the $(\text{Nb,Ta})_2\text{CN}$ phase and $(\text{Mo,W})(\text{Nb,Ta})\text{N}$ phase was determined to be $[100]_{(\text{Mo,W})(\text{Nb,Ta})\text{N}} \parallel [110]_{(\text{Nb,Ta})_2\text{CN}}$, $(004)_{(\text{Mo,W})(\text{Nb,Ta})\text{N}} \parallel (002)_{(\text{Nb,Ta})_2\text{CN}}$.

Another orientation relationship was identified between $(\text{Nb,Ta})_2\text{CN}$ and the HEA BCC grains as shown in Fig. 8. In this case a coherent $(\text{Nb,Ta})_2\text{CN}$ grain is protruding from the outside of parent HEA BCC grain towards the center the HEA BCC grain. The orientation of the BCC grain is down the [100] crystallographic direction while the $(\text{Nb,Ta})_2\text{CN}$ is oriented down the [110] direction. Therefore, the crystallographic relationship between the HEA BCC phase and the TaN phase was determined to be $[100]_{\text{HEA}} \parallel$

$[110]_{(\text{Nb,Ta})_2\text{CN}}$, $(002)_{\text{HEA}} \parallel (2-20)_{(\text{Nb,Ta})_2\text{CN}}$. A unique feature of the $[100]_{\text{HEA}} \parallel [110]_{(\text{Nb,Ta})_2\text{CN}}$ orientation relationship is that there was a rotation of $\sim 2^\circ$ between the HEA BCC regions on either side of the growing $(\text{Nb,Ta})_2\text{CN}$ grain.

Another comparison of grains containing the layered structure after various annealing treatments can be seen in Fig. 9. A comparison of the layered lattice structure after annealing at 1000°C between 1 and 1000 hours indicates that the $(\text{Mo,W})(\text{Nb,Ta})\text{N}$ phase experienced growth, while maintaining its orientation relationship, resulting in an increased layered lattice to $(\text{Nb,Ta})_2\text{CN}$ volume fraction ratio. This result indicated that the overall volume fraction of the layered lattice structure increased with increased annealing

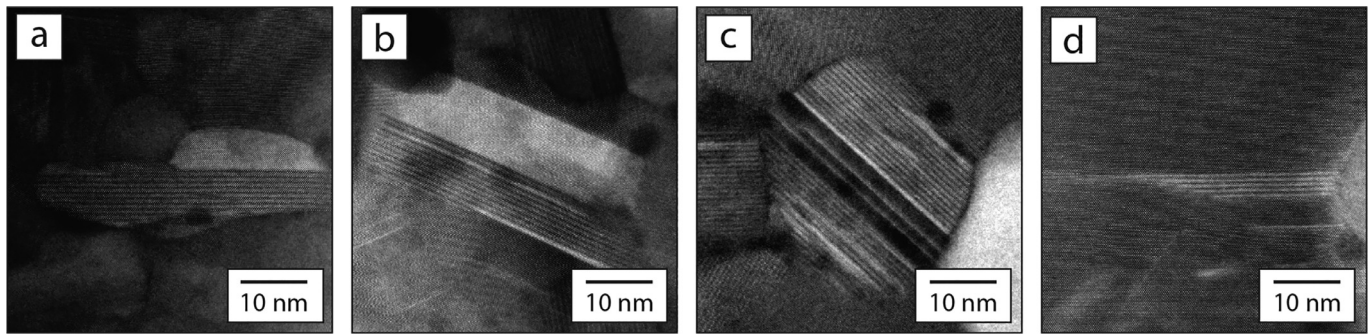


Fig. 9. STEM-HAADF images of grains containing the layered lattice structure after annealing HEA-N₂ at (a) 1000°C for 1 hour, (b) 1000°C for 100 hours, (c) 1000°C for 1000 hours, and (d) 1200°C for 100 hours.

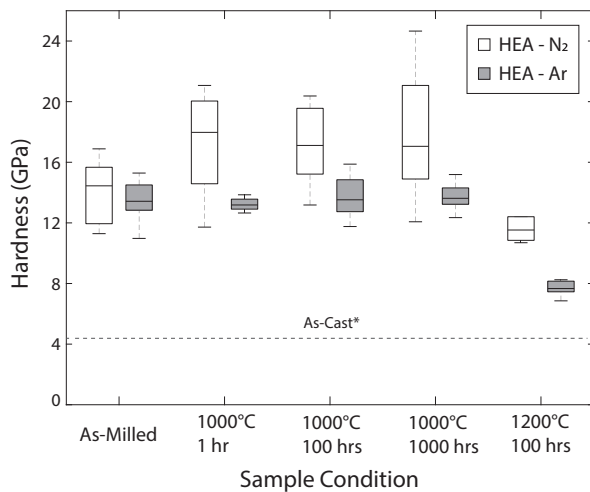


Fig. 10. Vicker's microhardness measurements for HEA-N₂ (white) and HEA-Ar (grey) at each annealing condition. A dashed horizontal line represents the reported hardness for as-cast MoNbTaW [1].

time. Furthermore, even after long-term annealing treatments of 100 hours (5b) and 1000 hours (5c) at 1000°C, the layered lattice structure was prevalent indicating the structure is thermally stable up to 1000°C for 1000 hours. In contrast, annealing at 1200°C for 100 hours resulted in a microstructure where the layered lattice structure was nearly non-existent. A grain containing a reduced amount of the layered lattice structure in HEA-N₂ annealed at 1200°C for 100 hours can be seen in Fig. 5d.

3.4. Effects of Nitride Formation on Hardness Mechanisms

The hardness values for the as-milled and annealed HEA-N₂ and HEA-Ar alloys are shown in Fig. 10. The horizontal line in each data set represents the median, the top and bottom limits of the boxes represent the 25th and 75th percentiles, and the caps represent the minimum and maximum values. The median hardness of as-milled HEA-N₂ (14.1 GPa) and HEA-Ar (13.4 GPa) were, on average, 300% harder than as-cast MoNbTaW (4.5 GPa). There were also measured differences in hardness upon annealing at 1000°C between HEA-N₂ and HEA-Ar; annealing HEA-N₂ consistently increased hardness up to 26% as compared to the as-milled condition, whereas HEA-Ar did not experience an increase in hardness. However, upon annealing at 1200°C for 100 hours, both alloys experienced a drop in hardness by about 50%. Considering similarities in grain size, this direct comparison of the HEA-N₂ and HEA-Ar indicates that

nitride formation significantly improved the hardness in the annealed HEA-N₂ alloys.

Considering that HEA-N₂ after annealing at 1000°C for 1000 hours exhibited the maximum hardness amongst all the alloys, a thin section was extracted underneath an indent to evaluate deformation mechanisms. Fig. 11a shows the thin section under the indent and Fig. 11b demonstrates that separate deformation mechanisms were observed for the HEA BCC phase and the complex nitride MoTa₂N. Most notably, a series of parallel planar defects (marked by black arrows in Fig. 11b) were observed in (Mo,W)(Nb,Ta)N while multiple dislocations were observed in the HEA BCC phase (marked by the dashed yellow circles). Fig. 11c shows an atomic-resolution STEM-HAADF image of a planar defect observed in the (Mo,W)(Nb,Ta)N nitride and Fig. 11d shows an inverse fast Fourier transform (IFFT) of the same region in Fig. 11c. The IFFT was reconstructed using the (0-12) reflections. The yellow arrows mark the continuous shift in atomic planes. Interestingly, as shown by Fig. 11c and d, the coherent phase boundary between the HEA and (Mo,W)(Nb,Ta)N phases acted as a barrier, stopping the planar defect (or shear plane) in its path. This behavior indicates that the shear mechanism is indeed unique to the (Mo,W)(Nb,Ta)N phase despite the coherency between the two phases and furthermore suggests that the coherent phase boundaries may play a significant role in strengthening. In contrast, Fig. 11e shows a STEM-HAADF image of an edge dislocation in the HEA BCC grain, which is more clearly identifiable in the IFFT that was reconstructed using the (110) crystallographic plane as shown in Fig. 11f. Overall, based on these results, the deformation mechanisms were distinctly different between the two prominent phases in HEA-N₂.

4. Discussion

4.1. Complex Carbonitride and Nitride Formation

Two alloys in this work were synthesized in which one was intentionally designed to contain high amounts of N (HEA-N₂) and the other alloy (HEA-Ar) was designed to contain significantly less N. The goal was to strategically form coherent nitride phases that increase the overall hardness of the material. A direct comparison of the HEA-N₂ and HEA-Ar alloy compositions indicates that HEA-N₂ contained ~24 at% nitrogen while HEA-Ar contained ~1 at%. In addition, one complex carbonitride and one complex nitride precipitated in HEA-N₂ upon annealing and exhibited distinct orientation relationships with each other and the main HEA BCC phase. More specifically, (Nb,Ta)₂CN and (Mo,W)(Nb,Ta)N formed after annealing at 1000°C. The presence of the complex carbonitrides and nitrides, (Nb,Ta)₂CN and (Mo,W)(Nb,Ta)N, in HEA-N₂, coinciding with the absence of nitrides/carbonitrides in HEA-Ar, is directly attributed to the use of liquid nitrogen during milling. Overall, the

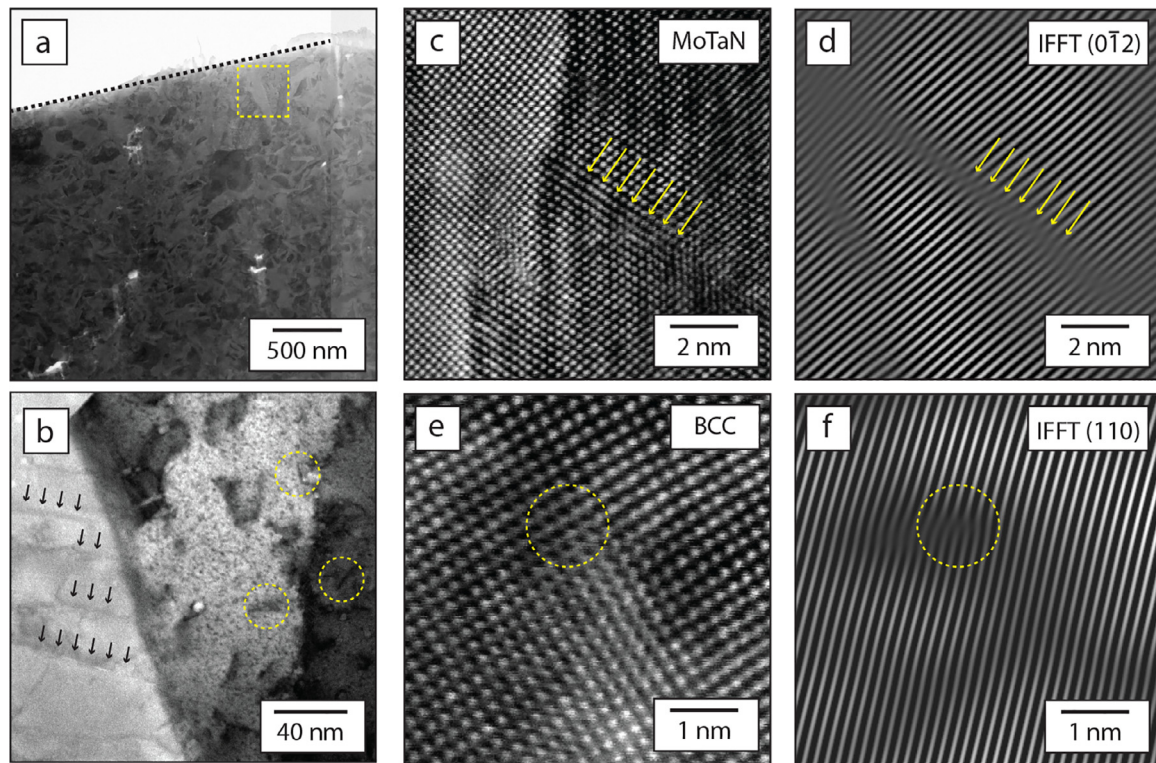


Fig. 11. (a) STEM-BF image of a cross-section under an indent from HEA-N₂ after annealing at 1000°C for 1000 hours. The indent is marked by the black dashed line. The yellow box marks the area shown in (b) that is a STEM-BF image of a region containing a coherent phase boundary between HEA BCC phase and (Mo,W)(Nb,Ta)N. (c) is a STEM-HAADF image of a shear plane in (Mo,W)(Nb,Ta)N where (d) is an inverted fast Fourier transform of the (0-12) plane. The yellow arrows in (c) and (d) mark the shear plane. (e) is a STEM-HAADF image of an edge dislocation in the HEA BCC phase where (f) is an inverted fast Fourier transform of the (110). The yellow circles in (e) and (f) mark the edge dislocation.

main objectives of this work, to develop coherent ceramic phases, were successfully accomplished.

The identified carbonitride and nitride in HEA-N₂ in this work are also unique as compared to other ceramic phases found in the literature. Both phases contained, in some concentration, all of the primary metallic elements wherein Nb and Ta shared atomic sites and Mo and W shared atomic sites. Moreover, even though the phase compositions are different than similar binary or ternary nitrides, there were insignificant differences in lattice parameters. Upon indexing FFTs as demonstrated in Fig. 5, the lattice parameter of (Mo,W)(Nb,Ta)N was measured to be $a = b = 3.096$ Å and $c = 7.843$ Å whereas the reported lattice parameter of MoTa₂N is $a = b = 3.095$ Å and $c = 7.866$ Å [52,53]. Likewise the lattice parameter of (Nb,Ta)₂CN was measured to be $a = b = c = 4.430$ Å whereas the reported lattice parameter of Nb₂CN is $a = b = c = 4.44$ Å [55].

According to a theoretical stability map of inorganic ternary metal nitrides, MoTa₂N has been previously observed as a metastable nitride that is stabilized under extreme conditions (i.e. high pressure) [56]. It has been experimentally verified that MoTa₂N can be synthesized at 1100°C in an atmosphere having an elevated nitrogen pressure equal to $P_{N_2} = 300$ bar [57]. Although atmospheric pressures encountered by the alloys in this experiment did not likely approach 300 bar, it is generally accepted that alloys formed by severe plastic deformation processes (i.e., mechanical alloying) encounter extraordinarily high amounts of strain. Therefore, it is suggested that high amounts of strain induced by mechanical alloying and the incorporation of nitrogen from milling with liquid nitrogen both enabled the formation of the relatively uncommon MoTa₂N-type phase. Overall, considering the extreme conditions necessary to form MoTa₂N (i.e., temperatures close to the melting point and pressures up to 1000 bar), this work has demon-

strated that cryomilling with liquid nitrogen is a viable and alternative method to synthesize ultrahard transition metal nitrides that are otherwise unstable.

4.2. (Mo,W)(Nb,Ta)N Thermal Stability

Although mostly amorphous in the as-milled condition, Fig. 2 and Fig. 7 indicate 1 hour annealing treatments of HEA-N₂ at 1000°C resulted in crystallization of 15–25 nm HEA BCC grains as well as the coherent, complex ceramic phases (Mo,W)(Nb,Ta)N and (Nb,Ta)₂CN. Considering that the (Mo,W)(Nb,Ta)N phase is potentially metastable, as previously discussed, long term annealing treatments up to 1000 hours at 1000°C were performed to assess the thermal stability and growth of (Mo,W)(Nb,Ta)N. It was observed that (Mo,W)(Nb,Ta)N likely nucleated on grain boundaries (i.e. heterogeneous nucleation) and propagated towards the center of primary HEA BCC or (Nb,Ta)₂CN grains via an edge growth mechanism analogous to twin growth in FCC metals (e.g. Cu). Furthermore, the layered (Mo,W)(Nb,Ta)N grains were stable upon annealing at 1000°C for 1000 hours. Thus, while previously considered to be metastable in other material systems, this work suggests that (Mo,W)(Nb,Ta)N is thermodynamically stable within nanostructured composites. The primary difference of the (Mo,W)(Nb,Ta)N phase after annealing for longer times, however, was that (Mo,W)(Nb,Ta)N completely transformed within primary grains after annealing for 1000 hours. Otherwise, in alloys annealed for shorter times, individual grains consisted of at least (Mo,W)(Nb,Ta)N and either the HEA BCC or (Nb,Ta)₂CN phases. Another critical result was that upon annealing at 1200°C for 100 hours, the ordered phase was no longer observed. Thus, this work has shown that the stability range of the (Mo,W)(Nb,Ta)N phase does not extend up to 1200°C.

4.3. Effect of (Mo,W)(Nb,Ta)N on Mechanical Properties

The microhardness of two alloys with the same intended bulk compositions ($\text{Mo}_{25}\text{Nb}_{25}\text{Ta}_{25}\text{W}_{25}$) were compared in this study. In the as-milled condition, both alloys showed ~350% improvements in hardness as compared to as-cast MoNbTaW, most likely due to the much smaller crystallite sizes in the as-milled alloys [1]. Upon annealing at 1000°C for 1 hour, an increase in the hardness was observed in both alloys and further annealing at 1000°C for extended times resulted in either an increase or no change in median hardness. A decrease in hardness in either alloy was not observed until significant grain growth occurred after annealing at 1200°C for 100 hours. Similar trends in hardness have been reported in nanocrystalline Ni-W and MoNbTaW whereby annealing an initially amorphous microstructure results in an initial increase in hardness. Most studies attribute this behavior to crystallization. Moreover, it is argued that further annealing induces grain growth resulting in corresponding decreases in hardness of up to 50% [21,58–62]. While similar trends in hardness were observed for HEA-N₂ and HEA-Ar, their hardness values differed considerably. More specifically, HEA-N₂ consistently exhibited a hardness 3–4 GPa greater than that of HEA-Ar. Overall, these hardness results highlight the beneficial effects of (Mo,W)(Nb,Ta)N and (Nb,Ta)₂CN on improving mechanical properties of MoNbTaW. Additional experiments aimed at evaluating the effect of complex ceramic phases on the ductility (i.e., tensile testing, nanoindentation, cantilever toughness testing, etc.) should be conducted to evaluate the potential of these alloys being used for extreme applications.

Extensive atomic-resolution characterization was performed on deformed materials to rationalize the increased hardness of HEA-N₂. Here, it was conclusively determined that HEA-N₂ was strengthened by coherent and complex nitride phases. From a macroscopic perspective, HEA-N₂ contained a much higher volume fraction of nitride phases which likely increased the overall hardness considering that refractory nitrides are harder than refractory metal phases. In this case, the intrinsic hardness of MoNbTaW, the primary HEA BCC phase in HEA-N₂, has been reported to be ~4.5 GPa [1,13]. While no experimental mechanical data could be found for the specific nitrides observed in this study, first-principles calculations estimate the hardness of MoTa₂N to be ~19 GPa, with the high hardness of MoTa₂N resulting from the highly directional covalent bonds between tantalum and nitrogen [53,56]. The specific strengthening mechanism of the complex (Mo,W)(Nb,Ta)N nitride observed in this work was that it deformed via a cleavage process, similar to many other brittle ceramics. Meanwhile the HEA BCC phase deformed via dislocation motion. The energy required to form a cleavage plane is most typically much higher than the energy required to form dislocations and thus, the improved hardness of HEA-N₂ is due to the presence of the complex nitride phase(s).

Of additional importance is the lattice mismatch between the ceramic and BCC phases, which results in a strengthening elastic strain energy at the phase boundaries. Analysis of the atomic resolution image in Fig. 11c provided that a misfit of $f = 0.113$ exists between the HEA phase and (Mo,W)(Nb,Ta)N lattices, respectively. This misfit was measured between the (0-12) shear plane in MoTa₂N and (110) in BCC. The magnitude of misfit measured here indicates that the interface between these two phases is either a highly-strained coherent interface or semi-coherent interface based on the criteria used to describe coherent interfaces in thin films [63]. Overall, in this particular example, the 0.113 planar misfit prohibits planar defects from translating between the HEA phase and (Mo,W)(Nb,Ta)N, which subsequently strengthens the alloy.

5. Conclusions

The main goal of this work was to develop refractory HEAs that contain coherent nitride phases in order to maximize

hardness. Two alloys with intended nominal compositions of $\text{Mo}_{25}\text{Nb}_{25}\text{Ta}_{25}\text{W}_{25}$ were synthesized via high energy cryogenic mechanical alloying. One alloy was milled with liquid nitrogen to incorporate N and promote secondary nitride phase formation as a mechanism to increase overall hardness. High temperature annealing treatments were completed followed by aberration-corrected scanning transmission electron microscopy (STEM) characterization and Vicker's microhardness indentation. Overall, it was determined that the optimized heat treatment was annealing at 1000°C for 1000 hours in Ar atmospheres to achieve a peak median hardness of 17.8 GPa, an almost 450% improvement compared to its cast relative. Atomic-resolution characterization determined the hardest alloy contained coherent (Nb,Ta)₂CN and (Mo,W)(Nb,Ta)N ceramic phases that deformed via a cleavage plane mechanism, as compared to dislocation-based deformation that is observed in HEA metallic phases. The overarching result of this work was the complete determination of processing-structure-property relationships pertaining to strategic incorporation of N into refractory HEAs, identification of heat treatments to precipitate and grow coherent ceramic phases, and establishment of annealing conditions to effectively age-harden nanocrystalline microstructures up to 24 GPa as a maximum observed hardness. Furthermore, this work demonstrates that cryogenic mechanical alloying is an adaptable technique to form complex ceramic phases that would otherwise require extreme conditions of temperature and pressure.

Data Availability

Data is available upon request.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.actamat.2021.116884.

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